Evaluation of large language models as expert systems for within-host malaria modelling



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Abstract

This study evaluated whether large language models can reliably conduct within-host malaria-modelling analyses. Eight language models attempted 55 short-answer and three synthetic-data computational tasks under single-response and agentic-workflow configurations. Language-model graders scored short answers; numerical checks and clean replay assessed submissions. Agentic versus single-response short-answer means were 73.9% versus 52.4%; 12 of 24 agentic submissions met execution, completeness and reproduction checks, versus 0 of 24. Safeguards caused seven of 55 Claude Opus 5 single responses to be empty and interrupted all 15 open-research attempts (five Fable 5 immediately; ten GPT-5.6 Sol later). One GPT-5.6 Sol attempt proposed a participant-varying effective male-contribution extension to a published *Plasmodium falciparum* within-host model before interruption. Independent reproduction in Stan, refitting and verification showed better internal sex-marker prediction than a matched Cao-derived global-allocation comparator: mean negative log predictive density was 4.65571 for the comparator versus 4.63580 for the extension, a gain of 0.01991 (95% participant-bootstrap interval 0.01173–0.02861), with improvement in 15 of 17 participants. Both fits passed the prespecified Markov chain Monte Carlo diagnostics. Overall, agentic workflows improved benchmark performance and showed potential to support substantive model development; however, their outputs required human verification, and safeguard-triggered interruptions substantially constrained open-ended research.

1 Introduction

Systems built around frontier large language models (LLMs) are beginning to move beyond synthesising established knowledge and towards contributing solutions to previously open research problems. Mathematics provides some of the clearest early evidence of this emerging capability. AlphaProof Nexus generated machine-checked Lean proofs for previously open Erdős problems [1]. More recently, a counterexample produced with Anthropic's Fable model refuted the long-standing Jacobian conjecture in dimensions three and above and was subsequently verified in Isabelle/HOL [2]. Together, these results provide early evidence that frontier LLM systems can contribute to open mathematical research.

The domains in which these systems appear to excel share an important property: success can be clearly defined and automatically checked. In mathematics, formal languages such as Lean can verify a proof; in software development, executable tests can check code against a specification. These checks create tight feedback loops in which candidate solutions can be tested and refined, allowing the system to explore the solution space efficiently [1].

Recent benchmarks have extended evaluation to data-driven science, research replication and health-related responses [3–6]. They indicate that tool-using agents can undertake components of scientific work, but also expose failures in execution, reproducibility and domain judgement. However, they do not establish whether language-model systems can integrate mechanistic biology, statistical inference and executable analysis in a specialist global health setting.

Within-host *Plasmodium falciparum* modelling provides a demanding test of this gap. Unlike formal proofs or software tests, infectious-disease models are evaluated against noisy, incomplete observations, and distinct biological mechanisms may produce observationally similar trajectories [7]. Transmission depends on mature male and female gametocytes, whose densities jointly influence mosquito infection, but sex ratios vary among infections and ecological settings [8, 9]. Immature gametocytes also sequester, hiding development until mature stage V re-enters circulation. Cao et al. modelled this latent pathway using asexual parasitaemia and piperaquine pharmacology. However, sexual commitment, sex-specific dynamics and mature-gametocyte clearance remain difficult to disentangle from peripheral measurements [10–12].

Against this background, we evaluated eight large language models on fixed malaria-

modelling tasks under two configurations: a single response without external tools and an agentic workflow with search, code execution, file access and subagent delegation (Figure 1). The benchmark combined short-answer questions with extended computational tasks requiring structured files, numerical results and written interpretation. This design tested both answer quality and whether a submission had reproducible results. All benchmark questions, reference answers and marking materials were kept non-public until evaluation of every model and configuration was complete, and were released only afterwards.

We therefore pursued two objectives. First, we compared the performance and reproducibility of eight large language models under single-response and agentic-workflow configurations. Second, in an exploratory open-research phase, we assessed whether frontier agentic systems could contribute to the development and independent validation of a new model of the partially observed *P. falciparum* sexual-stage pathway. This second objective extended the evaluation beyond benchmark performance to within-host gametocyte-model development.

2 Methods

2.1 Benchmark design

Benchmark conclusions depend on domain, task selection and construct definition [13, 14]. We therefore benchmarked candidate models on malaria-specific questions and computational tasks before selecting a model and agentic workflow for the open-research phase. Each model was evaluated under a single-response configuration and an agentic workflow (Figure 1).

The 58-item benchmark comprised 55 short-answer items and three extended computational modelling tasks. The placement supervisor supplied 38 question–answer pairs. We developed 20 additional items, informed by the current literature, to broaden the coverage: 12 conceptual questions, five fully specified multistep calculations and three extended tasks. The benchmark repository provides the complete short-answer questions, reference formulations and question-specific marking criteria as `build/full_short_answer_benchmark.pdf`, alongside canonical numerical inputs and grading code. The 55 short-answer items covered basic understanding (16), applied reasoning (18), synthesis and critique (16), and exact calculation

(5).

Each extended task comprised three elements, illustrated here by SELF-C07:

- **Dataset.** Each task supplied synthetic public inputs and withheld outcomes for evaluation. For SELF-C07, the public data comprised an asexual-parasite driver, mature female-gametocyte observations and forecast times; the corresponding gametocyte outcomes were held out.
- **Modelling task.** Each prompt specified a biological question. SELF-C07 required a non-negative compartment ODE linking the asexual trajectory to delayed mature gametocyte appearance and a comparison of alternative maturation structures.
- **Submission.** Each task required code, predictions with uncertainty, structured results and a write-up. SELF-C07 also required a trajectory file.

The benchmark revisited key principles at increasing levels of practical complexity. For example, SELF-B02 asked candidates to explain why circulating mature gametocytes depend on delayed stage-V inflow and clearance rather than autonomous growth. SELF-C07 then required this principle to be implemented in an executable delayed-maturation model [10, 15]. The remaining extended tasks covered parasite multiplication and mosquito infection, spanning asexual replication to onward transmission [8, 16–18]. Revisiting the same biology across task formats tested whether models could translate conceptual understanding into reproducible modelling work. Eight models were evaluated on every item under both configurations.

2.2 Evaluation and analysis

Each model received the same public task material in a single-response configuration and an agentic workflow. The former returned one answer without external tools; the latter built on Inspect AI's framework and the Deep Agent setup, which extends the ReAct pattern of interleaving reasoning, tool actions and observations with subagent delegation and structured planning [19–21]. It could search the web, delegate to subagents with isolated message contexts, inspect and edit files, execute Python and CmdStan analyses, and revise its output within a shared network-isolated environment. Single-response answers were allowed 768 output tokens,

whereas calculation and extended-task responses were allowed 12,000. All agentic runs permitted up to 50 messages and were capped at 500,000 cumulative tokens. Extended-task runs were limited to 1,200 seconds in the single-response configuration and 1,800 seconds in the agentic workflow. Submitted code was replayed and scored only after submission, separately from the agentic workflow.

Short answers were assessed against 114 question-specific marking criteria by three graders queried separately: GPT-5.5, Claude Haiku 4.5 and Gemini 3.5 Flash. The complete questions, reference answers and marking criteria are available in the repository PDF described above. Annex C.1 illustrates their application using supervisor-supplied item SUP-S11. The graders were drawn from the OpenAI, Anthropic and Google model families, following evidence that mixed-family panels reduce reliance on single-judge biases [22]. None of the grader models was included among the eight benchmark respondents. Each grader received the question, one valid reference formulation, the criteria and candidate text, but not the respondent or configuration label; this masking was intended to limit source-conditioned and self-enhancement bias [23]. Criterion coverage was averaged across valid graders. Separately, each grader classified the factual-error severity of each response, including false mathematics or biology, unsupported mechanisms and contradictions, as none, minor, major or central. The strictest valid classification capped the score at 1.00, 0.50, 0.25 or 0.00, respectively; the same procedure scored both configurations.

Extended tasks used the 100-point hybrid assessment shown in Figure 2. Deterministic checks of hidden predictions, numerical consistency and uncertainty outputs contributed 50 points; model-based assessment of statistical choices, biological interpretation, literature use and model criticism contributed the remaining 50. The approach adapted PaperBench's criterion-level scoring and clean execution-based verification [5]. Submitted code was replayed from public inputs in a clean environment with two CPU cores and a 600-second limit; missing, failed or non-reproducing outputs capped the total score at 50/100.

Health-specific criteria and evaluation of grader reliability drew on HealthBench [6], supplemented by checks for model-judge failure modes described in JudgeBench [24].

Scores were reported separately by task class; no aggregate score was calculated across

classes. Differences between configurations were calculated within model and summarised descriptively. Cost per target comprised provider-reported candidate-model usage, search-request counts and charges across all attempts [25].

2.3 Open-research task and agent setup

Open research comprised 15 attempts in modified versions of the agent harness: five targeting Fable 5 and ten using benchmark-selected GPT-5.6 Sol. Immediate rerouting away from Fable 5 or a later content block ending GPT-5.6 Sol work counted as a safeguard interruption. The configurations retained web search, file editing, Python and terminal execution in a shared workspace, while allowing the parent agent to spawn up to 64 concurrent same-model subagents in separate message contexts and synthesise their returned work.

The research task was specified in a 903-word instruction, reproduced in Supplementary Open-Research Prompt S1. The prompt was inspired by OpenAI's prompt used with GPT-5.6 Sol Ultra for its reported proof of the Cycle Double Cover Conjecture [26]. Our prompt asked the system to "discover, formulate, implement, and rigorously validate a genuinely step-change malaria model". It defined the primary research area as "the delayed, partially observed pathway from asexual parasites through sexual commitment and sequestered-but-developing immature gametocyte stages I–IV to circulating, non-replicating mature stage-V female and male gametocytes". A surviving candidate was required to outperform strong comparators using evidence not used in its design or selection.

The instructions directed the parent agent to begin with diverse, initially independent approaches, maintain a registry of substantive approach families, search the public research frontier and use adversarial subagents to test leakage, confounding, comparator fairness and numerical validity. The agent was permitted to return only after it had produced an executable model and independent evidence, reproduced the work and completed an adversarial audit; the prompt imposed a minimum research period of 12 hours.

2.4 Cao-derived hierarchical sex-marker model

Before safeguard interruption, one of the ten GPT-5.6 Sol Codex runs proposed adding a partially pooled participant-specific effective male contribution to the Cao framework. We subsequently reproduced the model and its results in Stan, fitting the extension alongside a matched global-allocation comparator and independently verifying the analysis using the procedure below. The analysis used a public dataset with coded identifiers from 17 malaria-naïve volunteers experimentally infected with *P. falciparum* 3D7 in controlled human malaria infection studies [10, 27, 28]. The dataset included total-parasite 18S qPCR measurements, female-marker Pfs25 and male-marker PfMGET qRT-PCR measurements, and participant-specific piperaquine concentration profiles. The abundance of Pfs25 and PfMGET transcripts was treated as a noisy proxy for gametocyte density rather than a direct cell count [29]. Cao et al. fitted a hierarchical, age-structured pharmacokinetic–pharmacodynamic model with 12 biological parameters per participant to the 18S data [10]. A constant fraction of asexual parasites entered a sexual-development pathway comprising four sequestered immature compartments and one circulating mature-gametocyte compartment. The original study compared the sex-marker measurements with model-derived gametocyte dynamics, but they did not contribute to its likelihood.

To model the longitudinal Pfs25 and PfMGET observations directly, the extension splits the common source $S_i(t)$ entering immature stage I into equal-kinetics female and male branches (Figure 3) and jointly fits both markers with assay-specific detection and noise terms. It allows the effective male contribution ρ_i to vary among participants through partial pooling:

$$\operatorname{logit}(\rho_i) = \operatorname{logit}(\rho) + \tau_\rho z_i, \qquad z_i \sim \mathcal{N}(0,1), \qquad \tau_\rho \sim \mathcal{N}^+(0,0.5), \tag{1}$$

$$S_{F,i}(t) = (1-\rho_i)S_i(t), \qquad S_{M,i}(t) = \rho_i S_i(t). \tag{2}$$

For participant i and sex-marker stream $s \in \{F,M\}$, with mature branch stock $G^V_{s,i}(t)$, the marker

observation model was

$$\mu_{s,i}(t) = \varepsilon + AG^{V}_{s,i}(t), \tag{3}$$

$$D_{s,i}(t) \sim \text{Bernoulli}\left(p_{s,i}(t)\right), \tag{4}$$

$$\text{logit}\, p_{s,i}(t) = \alpha_s + \beta_s \log\left\{1 + \frac{\mu_{s,i}(t)}{L}\right\}, \tag{5}$$

$$\log Y_{s,i}(t) \mid D_{s,i}(t) = 1 \sim \mathcal{N}\left(\log \mu_{s,i}(t), \sigma_s^2\right). \tag{6}$$

Here $D_{s,i}(t) = 1$ denotes detection of marker s for participant i at time t; ε, A and L are the latent floor, common amplitude and assay reference scale, while α_s, β_s and σ_s are marker-specific. In both newly fitted models, the global effective-male location ρ was assigned a $\text{Beta}(2,8)$ prior. The hierarchical-allocation extension added 17 shrunk latent effects and one heterogeneity scale, while conserving mass and retaining equal maturation and clearance between sexes. The global-allocation comparator fixed $\rho_i = \rho$ for every participant. Both models otherwise retained the Cao recurrence, biological-parameter priors, participant-specific PK inputs and censored 18S likelihood. This nesting makes the comparison a targeted test of participant heterogeneity in the effective marker contrast, rather than a comparison between different asexual-stage or pharmacological models.

For each participant, all Pfs25 and PfMGET measurements formed one assessment block. Block Pareto-smoothed importance sampling (PSIS) approximated a refit that omitted that participant's complete sex-marker block; across the 17 blocks, all 390 sex-marker observations were evaluated [30]. The target participant's full 18S and PK course remained conditioned upon. The analysis therefore measured internal participant-block sex-marker prediction conditional on complete 18S/PK, not prospective or external prediction. The primary model-comparison metric was mean NLPD, calculated by averaging observation-level NLPDs within each participant and sex and then weighting sexes and participants equally. Additional metrics included a participant-equal joint-block score and sex-specific participant-equal scores. Uncertainty was quantified using 50,000 participant-level bootstrap replicates.

Posterior qualification required maximum $\widehat{R} \leq 1.01$, bulk and tail effective sample sizes at least 400, no divergences or maximum-depth transitions, a minimum energy Bayesian fraction

of missing information of 0.30, and valid participant-block PSIS diagnostics [31]. Exact code, inputs, results and rebuild commands are provided in `sex-cao-model/`.

3 Results

3.1 Benchmark performance

Across eight models in the short-answer benchmarks, independently rounded means were 52.4% for single responses and 73.9% for agentic workflows, a difference of 21.5 percentage points. Mean differences were 19.9 points for basic understanding, 16.3 for applied reasoning, 26.0 for synthesis and critique, and 31.7 for exact calculation. Across the 440 model–question pairs, the agentic workflow scored higher in 188, tied in 198 and lower in 54. Model differences were heterogeneous (Figure 4). In the single-response condition, the Claude Opus 5 route also returned seven empty answers among the 55 questions, each with a content filter stop reason, meaning that generation was halted before any answer text was returned; no such stop occurred in its agentic condition. Domain-specific differences, cost–performance Pareto-frontier analysis and operational accounting are reported in Supplementary Figure S2, Supplementary Figure S3 and Supplementary Table S4.

3.2 Reliability, efficiency and workflow choice in extended tasks

All 48 extended-task submissions received a numerical score, but none of the 24 single-response submissions executed successfully. Mean scores for parasite multiplication were 6.25/100 and 64.11/100, for delayed gametocyte maturation 1.46/100 and 37.50/100, and for mosquito infection 8.54/100 and 52.80/100, under the single-response configuration and agentic workflow, respectively. Of 24 agentic workflow submissions, 16 executed, 13 contained every required non-empty file and 13 reproduced designated outputs; 12 met all three criteria. The corresponding counts by task were 6/8, 2/8 and 4/8.

GPT-5.6 Sol was the best-performing model under the agentic workflow, with a short-answer mean of 95.1% and an extended-task mean of 94.1/100. Its three extended-task scores were 87.4, 100.0 and 95.0.

In its highest-scoring extended-task response, SELF-C07, GPT-5.6 Sol represented immature gametocyte development using a five-stage Erlang transit model, fitted a left-censored lognormal observation model and favoured this structure over a single-stage alternative by AIC.

3.3 Cao-derived hierarchical sex-marker model results

One of the 15 open-research attempts produced a substantive modelling proposal: the Cao-derived hierarchical sex-marker extension described in Section 2.4. Although it did not meet the step-change promotion standard, independent reproduction and refitting established the predictive comparison reported below.

Primary participant- and sex-equal NLPD was 4.656 for the global-allocation comparator and 4.636 for the hierarchical-allocation extension. The comparator-minus-extension gain was 0.020 nats per observation (participant-bootstrap 95% interval 0.012–0.029), with the extension improving predictive performance for 15/17 volunteers (Figures 5–6). All participant-block PSIS diagnostics were valid: maximum Pareto k was 0.292 for the comparator and 0.318 for the extension.

Secondary analyses were concordant. The participant-equal joint-block gain was 0.014 nats (95% interval 0.007–0.021), with 13/17 participant contrasts favouring the extension. The female-marker gain was 0.006 (-0.003–0.016; 10/17), whereas the male-marker gain was 0.034 (0.019–0.051; 16/17). The larger and more consistent PfMGET gain matched the extension's target, but the female interval included zero.

The extension's population-location ρ had a posterior median of 0.244 (95% interval 0.189–0.306), and τ_ρ had a median of 0.385 (0.062–0.727); participant ρ_i means ranged from 0.191 to 0.359. These are effective marker contributions, not cellular sex ratios.

Both posterior fits converged. For the global-allocation comparator and hierarchical-allocation extension, respectively, maximum $\widehat{R}$ values were 1.004 and 1.006, minimum bulk effective sample sizes were 1,484 and 759, minimum tail effective sample sizes were 2,770 and 1,339, and minimum energy Bayesian fractions of missing information were 0.713 and 0.752. Neither fit had a divergence or maximum-depth transition. Full-chain traces and participant intervals are shown in Supplementary Figure S5. The supported claim is therefore improved internal sex-marker

prediction conditional on complete 18S/PK relative to the matched Cao-derived global-allocation comparator.

4 Discussion

4.1 Interpreting the benchmark

The 21.5-percentage-point short-answer difference should be interpreted as the advantage of the complete configured system on information-rich tasks: only the agentic workflow could search, execute code, revise files and delegate. It does not demonstrate improved intrinsic scientific reasoning.

Multi-step symbolic reasoning can be brittle [32], while program-aided approaches delegate calculation to an interpreter [33]. Consistently, exact-calculation items produced the largest mean difference (31.7 percentage points), suggesting that execution and checking complemented the underlying model.

The more consequential finding was the agentic workflow's ability to convert information into auditable computational analyses. Twelve of 24 agentic submissions executed successfully, contained every required file and reproduced their designated outputs, compared with none of the 24 single responses. Among the highest-performing models, OpenAI's GPT-5.6 Sol met all three criteria on every extended task and scored 87.4, 100.0 and 95.0/100 for parasite multiplication, delayed gametocyte maturation and mosquito infection, respectively, giving a mean of 94.1/100.

4.2 Safeguards as a constraint on scientific use

Safeguard intervention was the main operational barrier in the open-research task. In the single-response benchmark, the Claude Opus 5 route returned seven empty outputs among 55 benign malaria-modelling questions, each with a content filter stop reason indicating that generation had been halted before any answer text was returned. Safeguards were more disruptive during the open-research task. All five attempts targeting Fable 5 were intercepted immediately and routed away from the requested model. All ten GPT-5.6 Sol attempts began work but eventually encountered content blocks; only one produced a substantive candidate before interruption.

The providers describe a genuine dual-use problem. Anthropic states that Fable 5 was deliberately released with broad classifiers and that most biology and chemistry requests are handled by a less capable fallback model while the company works to reduce false positives [34]. OpenAI similarly documents real-time biology and cyber-misuse classifiers for GPT-5.6 and acknowledges that they can block legitimate work where beneficial and harmful activities initially appear similar [35]. The safeguards are intended to reduce harmful use; nevertheless, false-positive refusals imposed a substantive performance cost and constrained several lines of research enquiry.

Across 16 model–harness configurations, a recent biology benchmark found refusal rates of 7–74% for legitimate tasks and 1–62% for concealed hazardous tasks [36]. Open-weight models under different safeguard regimes would provide an informative comparison but were outside the available budget.

4.3 Interpreting the Cao-derived extension

This case illustrates how an agent originated a biologically informed model proposal that we subsequently reproduced and evaluated. It demonstrates agent contribution to a scientific workflow, not autonomous end-to-end discovery. Early source allocation is biologically plausible: sibling gametocytes from a committed schizont can share sex, and a transcriptional switch controls the subsequent male–female fate decision [37, 38]. However, the present model cannot locate that mechanism. With zero initial sex stocks and equal linear female and male kinetics, $F_i(t) = (1-\rho_i)G_i(t)$ and $M_i(t) = \rho_i G_i(t)$; the same likelihood results if the normalised contrast is placed in the terminal assay mapping. Consequently, ρ_i is an effective marker contribution that can absorb sex-specific transcript yield, assay recovery or omitted kinetics, not a demonstrated cellular sex ratio [29].

Deleting each target participant's complete Pfs25/PfMGET block while retaining their 18S/PK course yielded a small, consistent advantage over the matched Cao-derived global-allocation comparator, especially for the male marker. This is stronger than in-sample fit but remains an internal, conditional comparison. Distinguishing allocation from observation would require direct sex-cell calibration, participant-by-assay controls or sex-specific intermediate-stage

or kinetic information; broader superiority would additionally require untouched external data. A decisive follow-up would calibrate transcript abundance against sex-specific cell counts and test whether the estimated participant heterogeneity transfers to an independent cohort.

4.4 Limitations and next tests

Several design features limit generalisation. The eight models were purposively selected, with one accepted generation per model–item–configuration cell and only three extended tasks, so run-to-run variation was not estimated. Provider routes, versions and classifiers may change, and public reference material may have appeared in training data. The short-answer graders were not calibrated against independently double-marked expert answers, while the synthetic extended tasks tested implementation under known assumptions rather than clinical validity. The 20 items we developed were literature-grounded but not independently validated.

Because providers optimise models for particular tool environments, model and harness are increasingly coupled [39, 40]. Future evaluations should use repeated generations, blinded expert marking, matched tool ablations and provider-native harness comparators. Access logging should distinguish refusals, classifier stops and fallbacks. Benign and hazardous controls could then distinguish model limitations from deployment constraints.

5 Conclusions

Agentic workflows achieved higher mean scores and delivered more replayable malaria-modelling submissions than single responses in this paired benchmark. The contrast was heterogeneous and bundled multiple resource differences, so it does not identify the effect of any one tool or establish general expert-system competence. Repeated generations, expert-calibrated marking and matched workflow ablations are needed before the findings can be generalised beyond the evaluated models and tasks. Before safeguard interruption, one Codex run also proposed a Cao-derived hierarchical sex-marker candidate. Independent reproduction and evaluation found a small improvement in internal participant-block sex-marker prediction conditional on complete 18S/PK versus a matched Cao-derived global-allocation comparator, and both models met the

prespecified MCMC diagnostic thresholds. This does not establish superiority to the published Cao model, external or prospective generalisation, or a causal sex-allocation mechanism.

List of abbreviations

AIC, Akaike information criterion; LLM, large language model; MCMC, Markov chain Monte Carlo; NLPD, negative log predictive density; ODE, ordinary differential equation; PD, pharmacodynamics; PfMGET, *P. falciparum* male gametocyte-enriched transcript; PK, pharmacokinetics; PSIS, Pareto-smoothed importance sampling; qPCR, quantitative polymerase chain reaction; qRT-PCR, quantitative reverse-transcription polymerase chain reaction.

Declarations

Author contributions

- Conceptualisation: the author
- Data curation: the author
- Formal analysis: the author
- Investigation: the author
- Methodology: the author
- Software: the author
- Validation: the author
- Visualisation: the author
- Writing – original draft: the author
- Writing – review and editing: the author

Competing interests

The author declares that no competing interests exist.

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Data availability

The benchmark repository contains canonical benchmark records, a complete rendered bank at `build/full_short_answer_benchmark.pdf`, and the derived, participant-coded Stan input with attribution. Cao source data and code are available under CC BY 4.0 [28].

Code availability

Code, packaged derived data and documentation are available from the benchmark repository at https://github.com/mghthesis2026/thesis_code; the README describes their contents and structure.

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A Figures

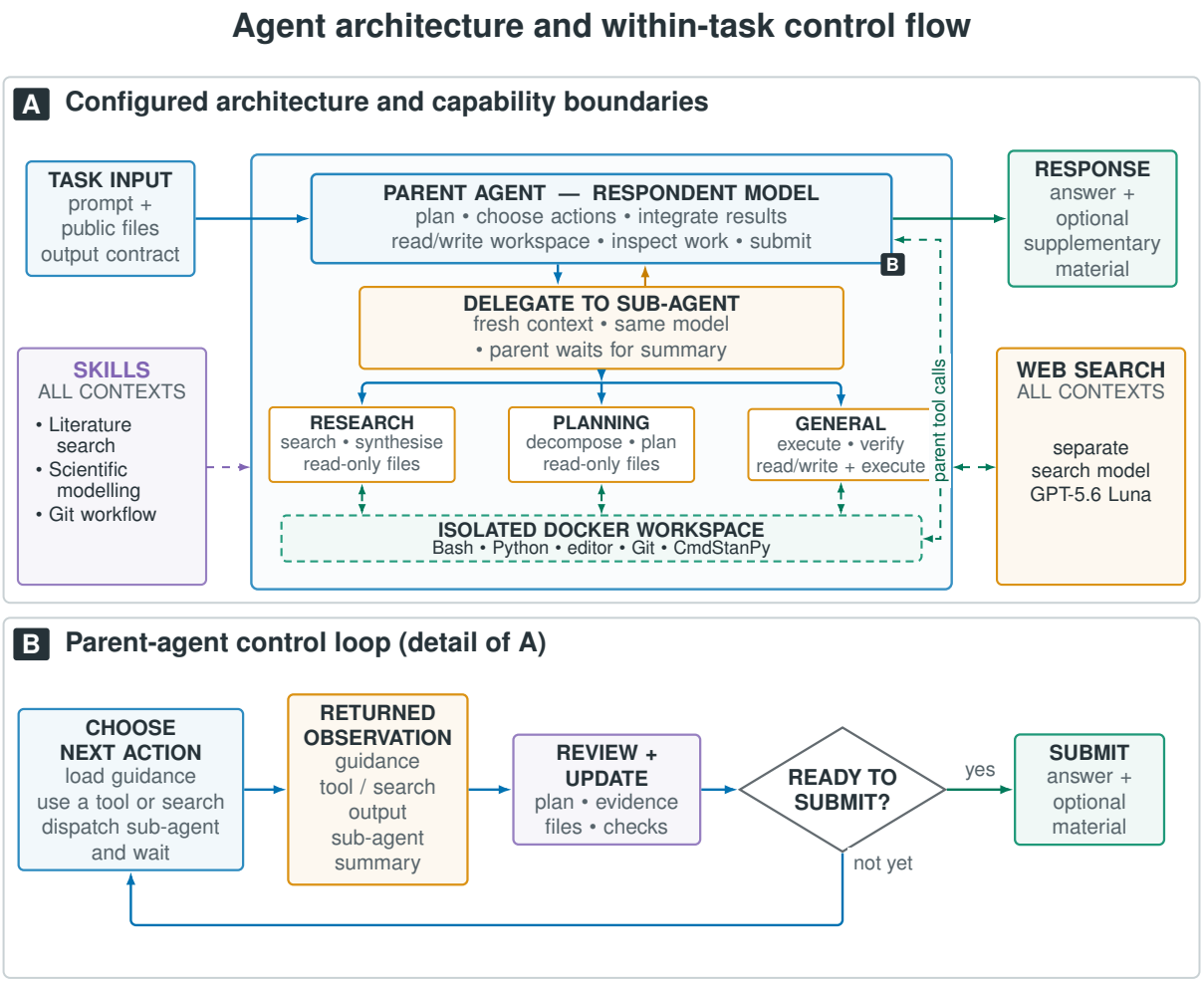


Figure 1: Agentic workflow architecture and dynamic within-task control flow. (A) A parent agent could delegate to a research, planning or general subagent. Each subagent used the same language model, began from a self-contained brief in a fresh message context and ran its own action–observation loop. All agent contexts could load three domain-guidance skills and use logged web search. Research and planning workers had read-only file interfaces; the parent and general worker could edit files and execute tools in one shared, network-isolated Docker workspace. The parent resumed execution when a subagent completed, and only the final summary returned to it. (B) On each turn, the parent could plan, load guidance, act in the workspace, search or delegate, then inspect the returned observation or subagent summary and either continue or submit.

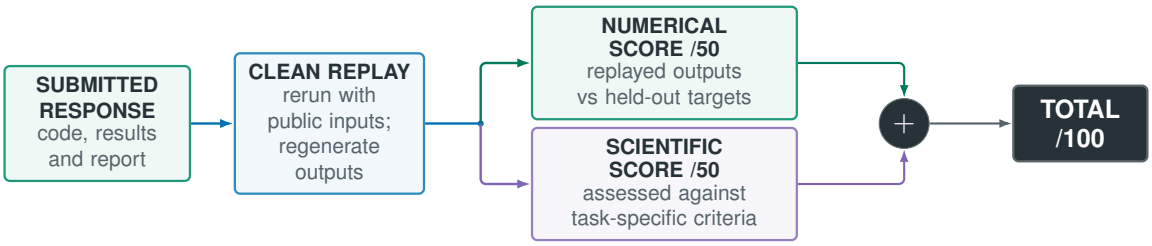


Figure 2: Evaluation workflow for extended computational modelling tasks. Clean replay supported numerical scoring, while the submitted methods and interpretation were assessed separately. Missing, failed or non-reproducing submissions capped the total at 50/100.

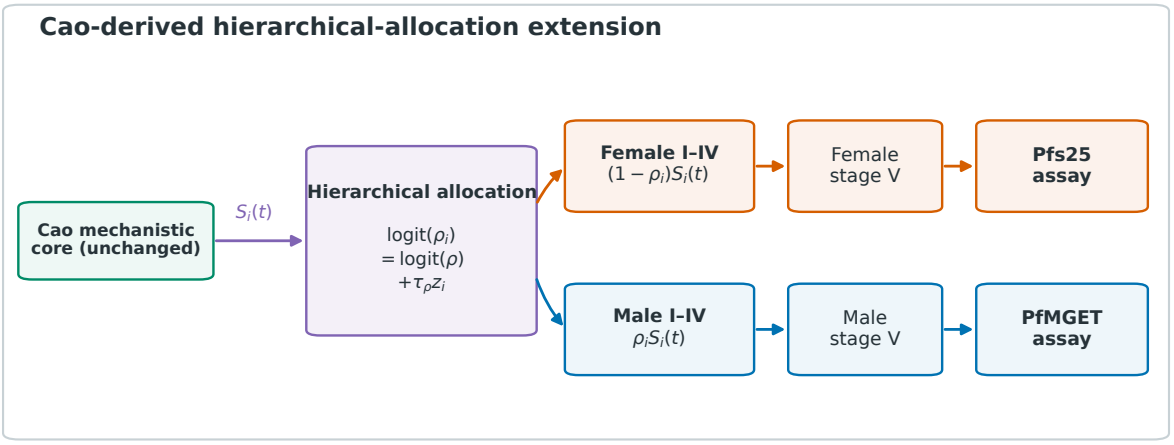


Figure 3: Cao-derived effective-sex-contrast extension. The published Cao upstream recurrence—PK/PD-modified asexual replication and sexual commitment—is unchanged and supplies the source $S_i(t)$ entering immature stage I. The hierarchical-allocation extension jointly models the sex markers and uses $\text{logit}(\rho_i) = \text{logit}(\rho) + \tau_\rho z_i$. Female and male terms sum exactly to the unchanged source, and all other model components retain matched maturation and clearance. With zero initial sex stocks and equal linear branch kinetics, source placement is observationally equivalent to a normalised participant-specific terminal assay-sex contrast. The resulting ρ_i is a marker-informed effective contribution, not a direct cellular sex ratio. The matched global-allocation comparator, retained in the comparative analyses but not diagrammed here, fixes $\rho_i = \rho$.

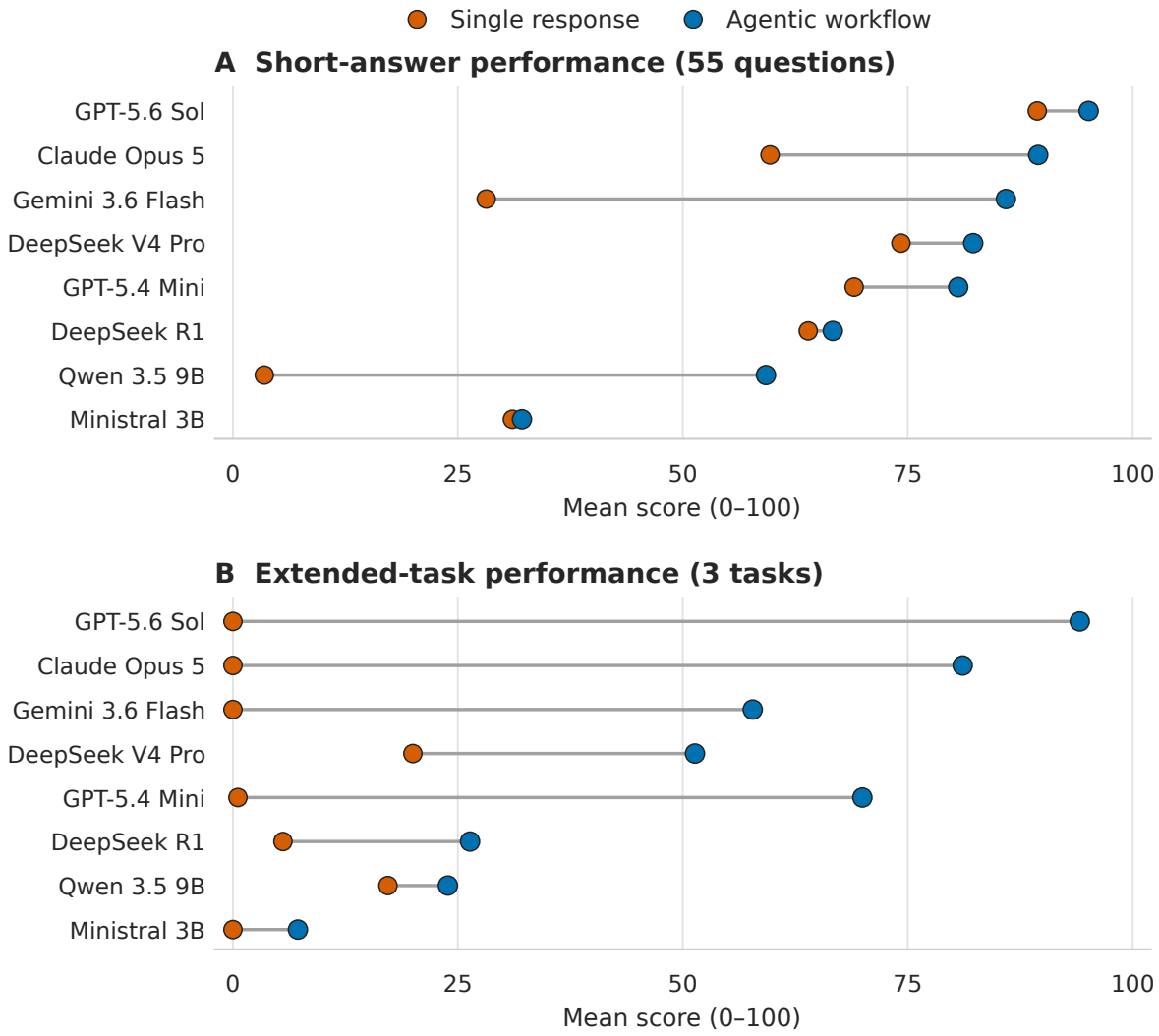


Figure 4: Benchmark performance. Each row pairs a model's mean score under the single-response and agentic-workflow configurations. Short-answer means cover 55 questions; extended-task means cover three tasks. Connecting lines identify paired configurations and do not isolate the effect of any one workflow component.

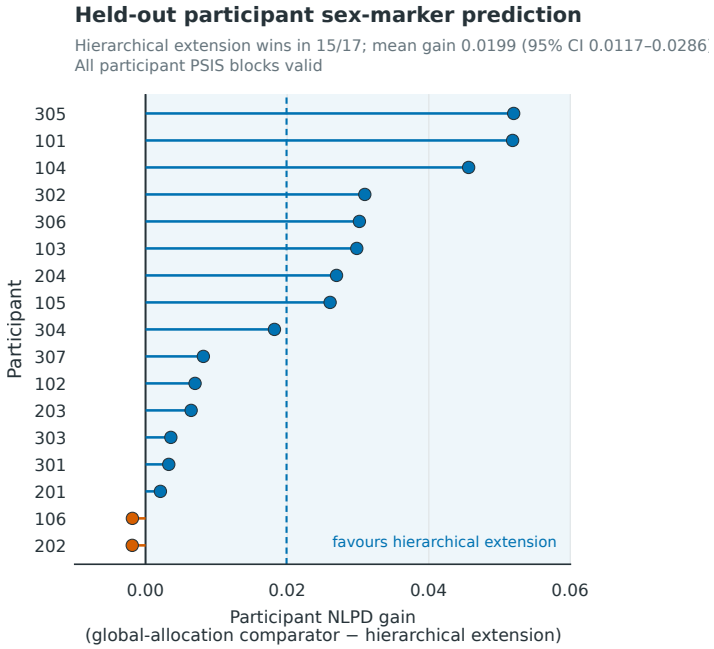


Figure 5: Participant-level internal sex-marker prediction gains. Positive comparator-minus-extension NLPD favours the hierarchical-allocation extension; 15/17 participant contrasts were positive, and the dashed line is their mean. Participant-block PSIS approximately removed each volunteer's complete sex-marker block while retaining the full 18S/PK course. This development-cohort conditional comparison is neither a prospective forecast nor external validation.

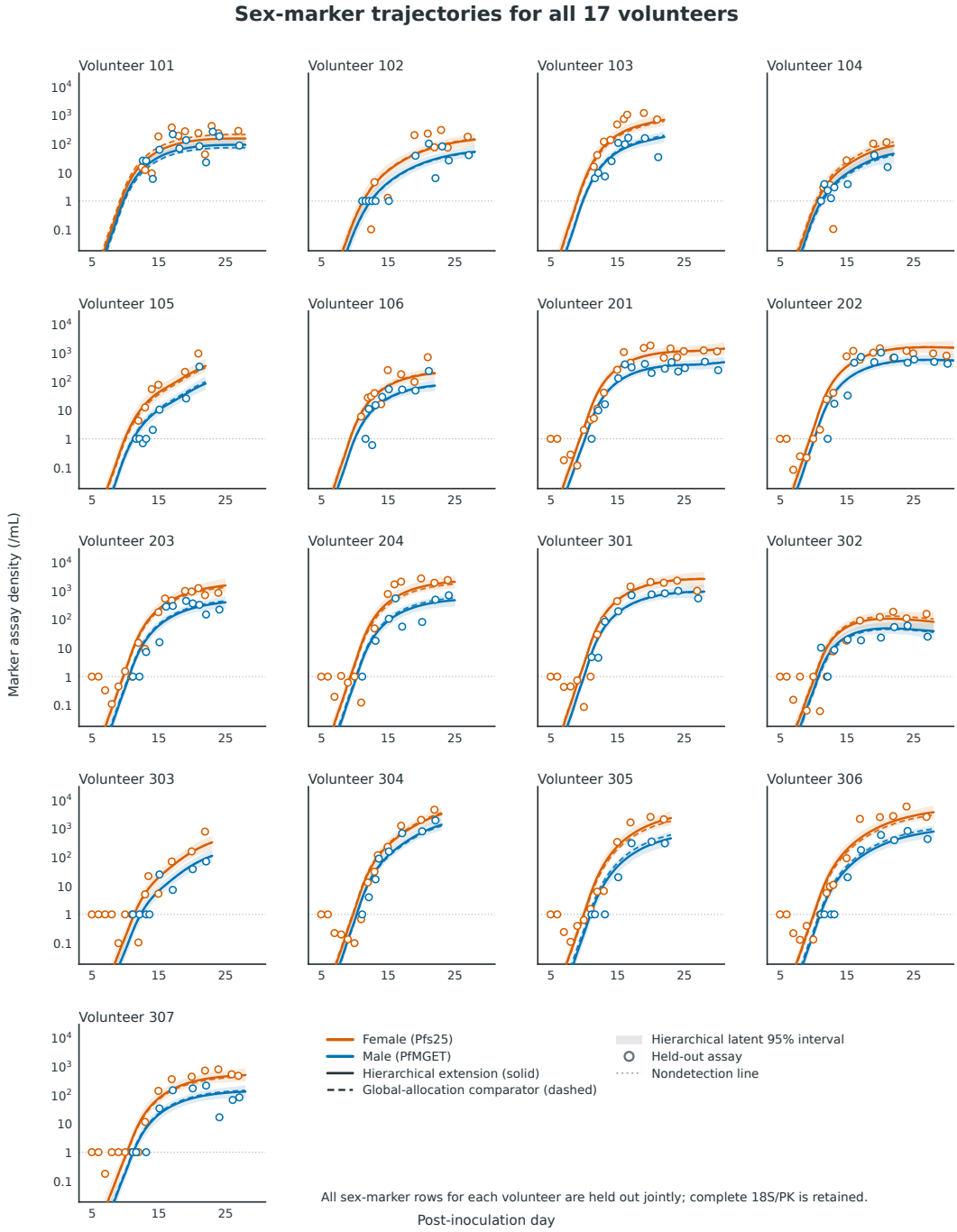


Figure 6: Sex-resolved analogue of Cao et al.'s Figure 3 [10]. Panels compare Pfs25 female-marker and PfMGET male-marker assay densities under the global-allocation comparator and hierarchical-allocation extension. As in Cao, observations are open circles; circles on the dotted one-per-mL line denote categorical nondetection rather than measured one-per-mL values. Participant-block PSIS approximately deleted each volunteer's complete sex-marker block while retaining complete 18S/PK. Solid curves and ribbons give extension case-deleted medians and 95% latent credible intervals; dashed curves give comparator medians. This same-cohort conditional diagnostic is neither prospective nor external validation.

B Supporting-information catalogue

S1 Open-Research Prompt (PDF). File: S1_open_research_prompt.pdf. Exact instruction supplied to the research agent.

S2 Figure: Domain-Stratified Benchmark Results (PDF). File: S2_domain_stratified_benchmark_results.pdf. Paired agentic-workflow minus single-response score differences by model and benchmark task group. Positive values favour the agentic workflow.

S3 Figure: Recorded Cost and Performance (PDF). File: S3_cost_performance.pdf. Paired performance and recorded candidate-model-plus-search expenditure by model, configuration and task class. Panels use different scoring systems and are not directly commensurate.

S4 Table: Benchmark Expenditure and Operational Audit (PDF). File: S4_expenditure_audit.pdf. All-attempt expenditure accounting and clean-execution, completeness and exact-replay outcomes for the extended agentic-workflow submissions.

S5 Figure: Cao-extension Posterior Diagnostics (PDF). File: S5_cao_posterior_diagnostics.pdf. Four-chain traces, participant posterior intervals and whole-model sampler summaries for the qualified global-allocation comparator and hierarchical-allocation extension fits.

C Short-answer marking example

C.1 Worked marking example

The following example uses SUP-S11, an item supplied by the placement supervisor.

Question. Can daily fluctuations in parasite count profiles alone be considered evidence for immune responses?

Reference answer. No. Daily fluctuations are primarily explained by parasite age structure and sequestration. Immune responses generally influence the longer-term trajectory (attenuation, plateau or decline) rather than the periodic oscillations themselves.

Marking criteria. The answer should:

1. reject daily fluctuations alone as evidence of an immune response;
2. attribute short-period oscillations primarily to parasite age structure and sequestration; and
3. distinguish these oscillations from longer-term attenuation, plateau or decline that could reflect immune control.